Assignment No.

Title: Data Visualization II

Objective:

1. Use the inbuilt dataset 'titanic' as used in the above problem. Plot a box plot for distribution of age with respect to each gender along with the information about whether they survived or not. (Column names: 'sex' and 'age')

Theory:-

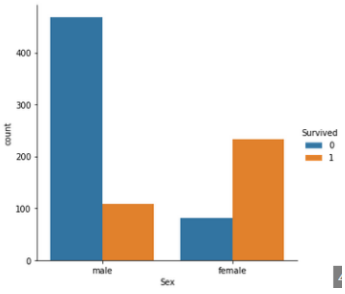
It’s hard to think of a professional industry that doesn’t benefit from making data more understandable. Every SYSTEM field benefits from understanding data and so d — o fields in government, finance, marketing, history, consumer goods, service industries, education, sports, and so on. While we’ll always wax poetically about data visualization (you’re on the Tableau website, after all) there are practical, real-life applications that are undeniable. And, since visualization is so prolific, it’s also one of the most useful professional skills to develop. The better you can convey your points visually, whether in a dashboard or a slide deck, the better you can leverage that information. The concept of the citizen data scientist is on the rise. Skill sets are changing to accommodate a data-driven world.

- Pairplot:
  By default, this function will create a grid of Axes such that each numeric variable in data will by shared across the y-axes across a single row and the x-axes across a single column.

```
sns.pairplot(penguins, hue="species")
```

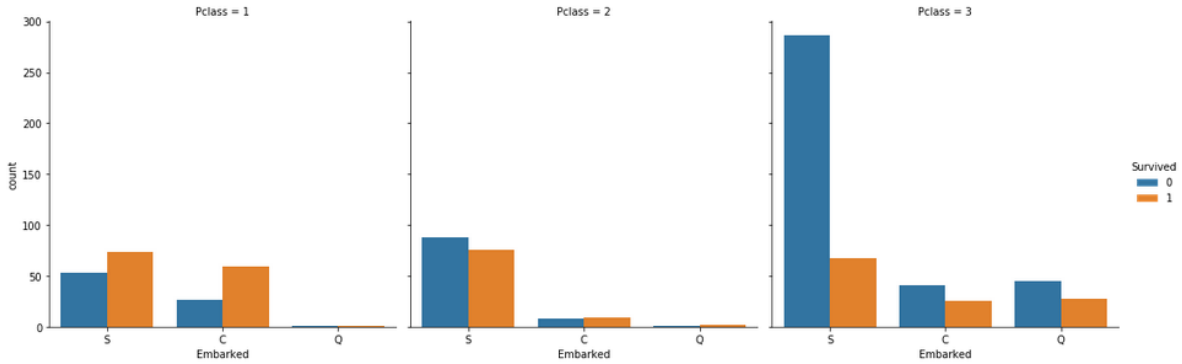
- Barplot
  A barplot (or barchart) is one of the most common types of graphic. It shows the relationship between a numeric and a categoric variable.

```
sns.barplot(data=df, x="island", y="body_mass_g")
```



- Count plot
  A count plot can be thought of as a histogram across a categorical, instead of quantitative, variable. The basic API and options are identical to those for barplot(), so you can compare counts across nested variables.

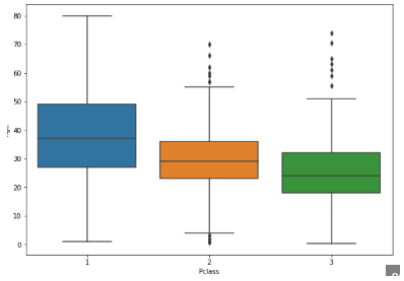
```
sns.countplot(x=df["class"])
```



- Box plot

  A box and whisker plot—also called a box plot—displays the five-number summary of a set of data. The five-number summary is the minimum, first quartile, median, third quartile, and maximum. In a box plot, we draw a box from the first quartile to the third quartile. A vertical line goes through the box at the median.

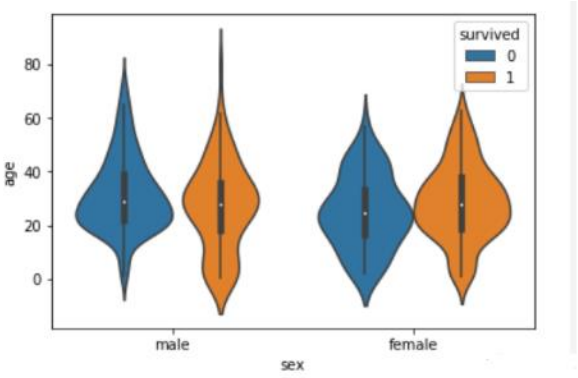
  sns.boxplot(x=df["age"])



- Volien plot

  A violin plot is a hybrid of a box plot and a kernel density plot, which shows peaks in the data. It is used to visualize the distribution of numerical data.

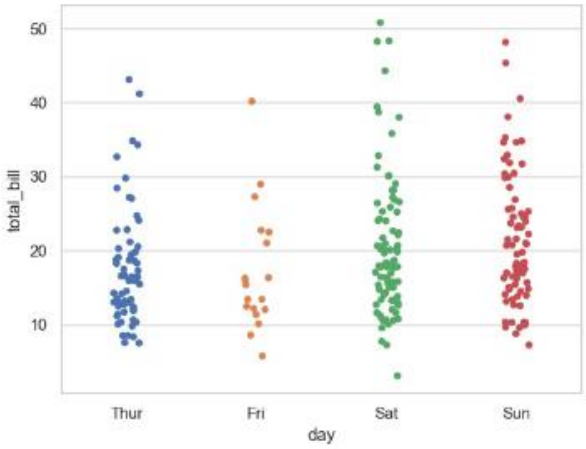
  sns.violinplot(x=df["age"])



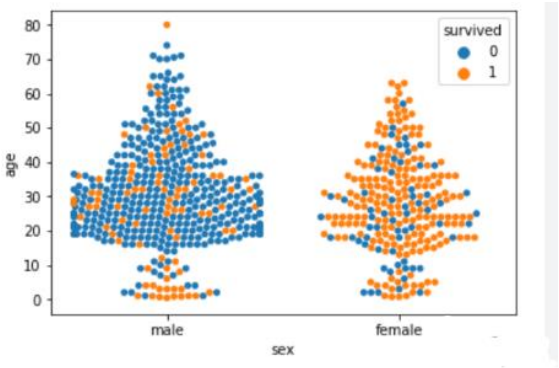
- Strip plot

A strip plot can be drawn on its own, but it is also a good complement to a box or violin plot in cases where you want to show all observations along with some representation of the underlying distribution.

sns.stripplot(data=tips, x="total_bill")



- Swarm plot
  This function is similar to stripplot(), but the points are adjusted (only along the categorical axis) so that they don't overlap. This gives a better representation of the distribution of values, but it does not scale well to large numbers of observations. This style of plot is sometimes called a "beeswarm".



Conclusion: Thus we have successfully Implemented Simple Data visualization techniques using Python on Titanic dataset.